

Enhancing interaction performance of soft pneumatic-networks grippers by skeleton topology optimization

CHEN ShiTong¹, WANG YuSheng¹, LI DeChen¹, CHEN FeiFei^{1,2*} & ZHU XiangYang^{1,2}

¹ Robotics Institute, School of Mechanical Engineering, Shanghai Jiao Tong University, Shanghai 200240, China;

² State Key Laboratory of Mechanical System and Vibration, Shanghai Jiao Tong University, Shanghai 200240, China

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The inherent compliance of soft materials imbues robots, generally referred to as soft robots, with particular advantages in producing adaptive and safe interactions. However, the mainstream design paradigms of soft robots have been focused on pursuing large free motions only, usually at the expense of greatly decreased stiffness, leading to limited capability of withstanding external loads in interactive scenarios. There is a pressing need to incorporate the interaction specifications at the design stage to embody soft robots with not only proper deformability but equally importantly, considerable stiffness to perform complex tasks in practical applications. Here, inspired by the dexterity of human hands, we propose a computational design framework for soft grippers with a focus on improving their interaction performance in power grasping or precision grasping mode. The design paradigm rests on attaching a relatively stiffer skeleton layer to the parametric pneumatic networks based actuator which is widely used due to the geometric advantage, and the skeleton layout is designed for customized interaction conditions by a level set based topology optimization approach. As expected, the optimized skeleton layouts exhibit specified structural features highly relevant to the predefined concentrated loads for precision grip or distributed loads for power grip, which physically implies the compromise between deformability and stiffness. Since the interaction forces are difficult to measure *in situ*, we devise power and precision grasping scenarios and evaluate the critical actuation pressure of the object's falling instead. The experiments qualitatively demonstrate the superiority of each specified design. This work represents an initial step toward the rational design for interaction in soft robots.

soft robotics, interaction, grippers, topology optimization

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1 Introduction

Interactions may often occur when a robot performs a task such as manipulation and grasping. Soft robots constructed from deformable materials have particular advantages in producing conformal and safe interactions, which are thus attractive for many scenarios [1,2]. For example, a soft gripper may gently grip fruits [3] and fragile objects [4], a soft wearable glove may assist a stroke patient with ease and

comfortableness in hand function rehabilitation [5], and a compliant flapping wing may adaptively reconfigure itself in dynamic air flows to attain a high lift-drag ratio (<https://www.festo.com/group/en/cms/13787.htm>). A soft-bodied robot literally has infinite degrees of freedom and thus is typically underactuated, but its inherent morphological compliance allows the soft body to deform continuously and interact with external environments, without significantly increasing the control complexity [6]. To this end, the incorporation of interactions at the design stage is of great importance to imbuing a soft robot with specialized mor-

*Corresponding author (email: ffchen@sjtu.edu.cn)

phology for target tasks.

Current mainstream design paradigms of soft robots have been focused on the free motion behaviors, such as bending and twisting. However, from a structure perspective, without a proper consideration of power performance, the stiffness of the robot is usually recklessly traded for producing large motions. A straightforward example is the widely used Pneumatic Networks (Pneu-Nets) based bending actuator [7]. Relying on fluctuated and distributed air channels, soft Pneu-Nets actuators attain remarkably asymmetric stiffness to generate large bending deformation upon pressurization, but the folded pneumatic chambers undesirably reduce the moment of inertia and weaken the bending stiffness of actuators to carry external loads from the environment. That is, the power performance of soft Pneu-Nets grippers is limited, and one has to resort to other remedies such as replacing the elastomer with stiffer materials (<https://www.softroboticsinc.com/products/>), or embedding stiff materials within the channel gaps [8]. However, these remedies would somehow in turn sacrifice conformal interaction characteristics.

The “optimal” compromise between deformability and interaction performance necessitates the concurrent consideration of them when designing a soft robot toward prescribed tasks. The interactions of soft robots with environments are generally spatiotemporally varying [9]. In this paper, we focus on the design of soft grippers, and the interaction conditions vary with the grip mode such as power grip and precision grip. More precisely, the interacting forces are concentrated for precision grip and distributed for power grip, with the force direction depending on the configuration of the gripper.

Continuous efforts have been made toward computational design of soft robots [1]. In particular, topology optimization has emerged as a powerful tool to automatically generate the design in soft robots, e.g., soft grippers actuated by cables [10], pneumatics [11], and ferromagnetic fields [12], but few work has effectively taken into account the interaction conditions in the design model. Zhang et al. [13] introduced a topology optimization method for the design of a bending actuator. The external force was modelled by a spring which generated a concentrated interaction force. By maintaining the output displacement constant with a spring model, Liu et al. [14] proposed a constant-force compliant gripper. However, the underlying assumption that the contact occurs at a fixed point does not suffice by noting an analog in nature that human hands demonstrate high dexterity for various contact conditions [15]. Human hands can adjust the gesture and stiffness to handle the object based on the dimension [16], shape and weight [17]. For example, to handle delicate objects of small size, a pinch grip increases the grasping precision. For objects of large size, one expects an increase in the contact area to deliver a wrapping grip, which favors the stability of grasping [18]. Chen et al. [10] investigated how

the topology optimized design depended on the interacting normal force and frictions, in the context of linear optimization, which is not directly applicable to soft gripper design.

In this work, we aim to develop an automatic computational design method of soft pneumatic grippers with desired interaction performance. We have previously proposed a design paradigm for soft pneumatic actuators by attaching a skeleton layer which is subject to topology optimization to an inner softer inflated chamber [19], with desired motions only as the design objective. Here, we further extend the work by generalizing the design domain to a class of Pneu-Nets actuators and more importantly, by considering the interacting forces throughout the optimization process. Specifically, we retain the geometry advantage of Pneu-Nets and further generalize it by parameterization. When designing the skeleton layout by topology optimization, to allow manipulating the design process in 2D space, a forward geometry mapping is constructed from a planar design space where we define the level-set function to describe the skeleton profile implicitly to the physical skeleton domain, and then an inverse design velocity mapping is constructed to transfer the sensitivity information. The level-set function is dynamically updated until convergence according to the gradient-based sensitivity analysis which reveals how the pursued motion behavior under interactions depends on the skeleton layout.

A design and postprocessing framework is proposed for manufacturing and prototyping of soft grippers. To shed light on the effect of interactions on the optimal design, we investigate three typical interaction cases of a soft finger, including (1) a concentrated force at the fingertip; (2) distributed forces on half of the finger interface; and (3) fully distributed forces. As expected, the optimal structural features of skeleton to a large extent depend on the interactions and they each automatically reach an adaptive compromise between proper deformability to conform to grasped objects and considerable stiffness to withstand external payloads. To experimentally validate the optimized design, we first demonstrate the effectiveness of incorporating skeleton layer by comparing with purely soft Pneu-Nets actuator without skeleton, and it is found that the skeleton layer well constrains the radial expansion and significantly enhances the bending motion. Further, to verify the superiority of design optimization for specified interaction conditions, we devise a gripping and falling test to qualitatively compare the grippers consisting of specialized skeletons by measuring the critical actuation pressure of objects' falling. We find that a gripper designed for power grip indeed performs better in practical power grasping tests and a gripper designed for pinch grip indeed performs better in practical precision grasping tests, which proves the superiority of skeleton design and the necessity of interaction consideration when designing soft robots.

2 System and model

2.1 Parametric pneumatic networks (PPNs)

The actuator consists of a bellows-shaped soft rubber and a stiffer skeleton layer attached seamlessly to the exterior surface of rubber. The inflated rubber encased inside remains unchanged while the exterior skeleton is allowed to change its topology to program the overall actuator's motion behavior. We parameterize the widely used pneumatic networks actuator, referred to as PPNs. The geometry of the PPNs is described by several independent parameters, as shown in Figure 1. Each cross-section profile consists of two semi-ellipses whose minor axes are equal to b , while the semi-ellipse in bottom half has the major axis a_2 . In order to form the bellows-shaped structure, the major axis in top half varies along z axis and is expressed by $a_1 \cdot f(z)$, where $f(z)$ is a periodic factor described by a trigonometric function:

$$f(z) = (s + 1) / 2 + (1 - s) / 2 \sin(2\pi z / T), \quad (1)$$

where s controls the amplitude and T is the period of bellows. The actuator is of length l and uniform thickness t^r . Hence, the geometry can be described uniquely by a set of parameters $(a_1, a_2, b, s, t^r, T, l)$.

2.2 Reinforced skeleton

The thickness of the reinforced skeleton layer is set to be uniform. We map the geometry of the skeleton layer into a rectangular design domain which is described by a dynamic level set function. Level sets have been widely used to capture and track the topology of freeform shapes, and have been compatibly integrated with topology optimization methods [20]. The boundaries of the geometry can be implicitly represented by the zero level-set as demonstrated in Figure 2. The material distribution refers to the value of the level set function: entity $\Omega / \partial\Omega$, boundaries $\partial\Omega$ and voids D/Ω , formulated by

$$\begin{cases} \phi(\mathbf{x}, t) > 0, \forall \mathbf{x} \in \Omega / \partial\Omega, \\ \phi(\mathbf{x}, t) = 0, \forall \mathbf{x} \in \partial\Omega, \\ \phi(\mathbf{x}, t) < 0, \forall \mathbf{x} \in D/\Omega, \end{cases} \quad (2)$$

where t is introduced as the pseudo-time to describe the evolution of the boundaries governed by the Hamilton-Jacobian equation:

$$\frac{\partial \phi(\mathbf{x}, t)}{\partial t} = -|\nabla \phi(\mathbf{x}, t)| V_n^r, \quad (3)$$

where V_n^r refers to the normal evolution velocity on the boundary: $\mathbf{x} \in \partial\Omega$ in planar design domain. Assuming the design domain is of width w and height h , the mapping relation from the planar design domain $P : (p, q) \in \mathbb{E}^2$ to the physical domain $\bar{P} : (x, y, z) \in \mathbb{E}^3$ is formulated by

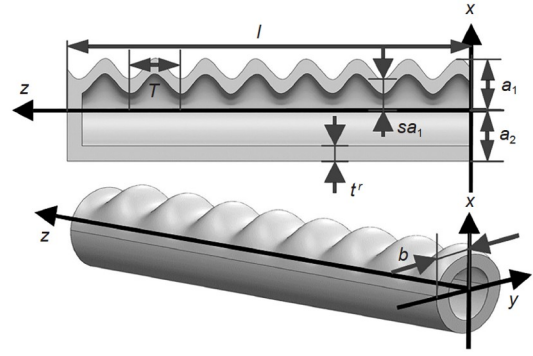


Figure 1 Geometric parameters of PPNs.

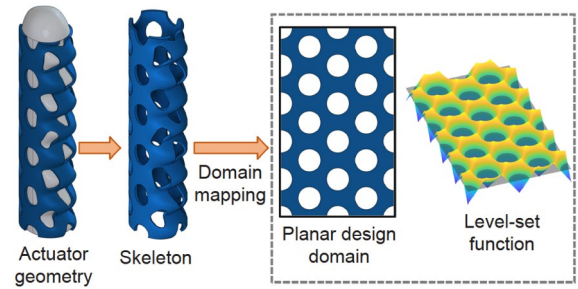


Figure 2 (Color online) Skeleton layout representation by mapping the physical domain to planar design domain where level sets are defined.

$$\text{for } x \geq 0: \begin{cases} x = a_1 f(z) \cdot \cos(\pi p / w + 3\pi / 2), \\ y = b \cdot \sin(\pi p / w + 3\pi / 2), \\ z = lq / h, \end{cases} \quad (4)$$

$$\text{for } x < 0: \begin{cases} x = a_2 \cdot \cos(\pi p / w + \pi / 2), \\ y = b \cdot \sin(\pi p / w + \pi / 2), \\ z = lq / h. \end{cases} \quad (5)$$

2.3 Mapping of design velocity

Since the skeleton geometry subject to optimization rests on the 2D space, the design velocity defined in the physical 3D domain should be mapped inversely to the 2D design domain. The evolution velocity in 3D space denoted by V_n^c lies in the local tangent plane of skeleton surface and is normal to the skeleton boundaries.

For an arbitrary vector $\mathbf{a}^r$ in 2D design domain, the corresponding forward mapping is constructed by $\mathbf{a}^c = \mathbf{J}\mathbf{a}^r$, where $\mathbf{a}^c$ is the vector in 3D space after mapping and $\mathbf{J}$ is the Jacobian matrix according to eqs. (4) and (5). The inverse mapping of V_n^r is expressed by

$$V_n^r = V_n^c \mathbf{G} \mathbf{n}^c \cdot \mathbf{n}^r, \quad (6)$$

where $\mathbf{G}$ is the inverse of $\mathbf{J}$, $\mathbf{n}^c$ and $\mathbf{n}^r$ denote the unit direction vector of V_n^c and V_n^r , respectively.

3 Skeleton design by topology optimization

This section will formulate the skeleton geometry design for the soft Pneu-Nets grippers as a topology optimization problem and develop a design-manufacture work flow for executing grasping tasks.

3.1 Optimization model

Soft Pneu-Nets actuator exhibits large deformation upon pressurization. The interaction forces of interest that occur during grasping are modelled by distributed loads applied to the normal of the end-effector, in the context of deformed configurations. The interaction conditions are characterized by the contact area and the magnitude of the pressure which in practice depend on the specific grasping scenario. For instance, for objects of small size, one may expect a precision grasp in which the interaction load concentrates at the fingertip of the gripper. For objects of larger size and weight, power grasp is more effective and a large-area distribution along the actuator is expected [18].

We adopt the generalized Neo-Hookean model and linear elastic model to characterize the soft rubber and skeleton, respectively. To capture the geometry nonlinearity, the Green-Lagrange strain $\mathbf{E}$ and the work-conjugate second Piola-Kirchhoff stress Σ are used. With a quasi-static assumption, the state equation is formulated as follows:

$$a(\mathbf{u}, \mathbf{v}, \phi) = l(\mathbf{u}, \mathbf{v}), \quad \forall \mathbf{v} \in \mathbf{U}, \quad (7)$$

where $\mathbf{u}$ is the displacement field that belongs to the kinematically admissible displacement space $\mathbf{U}$, while $\mathbf{v}$ denotes the virtual displacement field. The variational structural form is

$$a(\mathbf{u}, \mathbf{v}, \phi) = \int_{\Omega_0^{\text{ske}}} \Sigma(\mathbf{u}) : \bar{\mathbf{E}}(\mathbf{u}, \mathbf{v}) d\Omega + \int_{\Omega_0^{\text{rub}}} \Sigma(\mathbf{u}) : \bar{\mathbf{E}}(\mathbf{u}, \mathbf{v}) d\Omega, \quad (8)$$

where Ω_0^{ske} and Ω_0^{rub} denote the undeformed region of the skeleton and PPNs, respectively. The variational load form is

$$l(\mathbf{u}, \mathbf{v}) = p \int_{\Gamma_0^{\text{rub}}} \mathbf{v} \cdot (\mathbf{m}_{\text{rub}} \cdot \mathbf{F}^{-1}) J d\Gamma + p^{\text{int}} \int_{\Gamma_0^{\text{int}}} \mathbf{v} \cdot (\mathbf{m}_{\text{rub}} \cdot \mathbf{F}^{-1}) J d\Gamma, \quad (9)$$

with Γ_0^{rub} the inner surface for pressurization load p , Γ_0^{int} the contact area of the actuator, $\mathbf{m}_{\text{rub}}$ the normal to the surface in the reference configuration, $\mathbf{F}$ the deformation gradient and J the determinant of $\mathbf{F}$. Here we assume that the interaction pressure p^{int} is constant.

For a soft gripper design, both conformability and stiffness are pursued. It is noted that, the conformal contact depends on not only the actuator but also the physical parameters of the objects. Since the gripper is expected to handle a wide

variety of objects varying in shape and size, the desired conformal shape cannot be quantitatively determined at the design stage, but instead we use a heuristic that the conformability is positively correlated to the bending capability of the soft finger. To this end, we represent the conformability of the actuator by the bending angle, and the optimization model is formulated by

$$\begin{aligned} &\text{minimize: } J_{\text{obj}} = -\theta^*, \\ &\text{subject to: } a(\mathbf{u}, \mathbf{v}, \phi) = l(\mathbf{u}, \mathbf{v}), \quad \forall \mathbf{v} \in \mathbf{U}, \end{aligned} \quad (10)$$

where θ^* is the bending angle of the end face in y axis as shown in Figure 3. Herein, possible interacting forces are considered in the mechanical model. That is, the actuator is expected to bend as much as possible while sustaining prescribed interacting forces.

By constructing the Lagrangian function, the constrained optimization model translates into an unconstrained problem. We carry out the shape sensibility analysis using the adjoint method to infer how the bending motion quantitatively depends on the skeleton layout. The normal velocities on the skeleton boundaries are

$$V_n^c(\mathbf{x}) = -\Sigma(\mathbf{u}) : \bar{\mathbf{E}}(\mathbf{u}, \mathbf{w}), \quad \forall \mathbf{x} \in \Gamma_0^{\text{ske}}, \quad (11)$$

with $\mathbf{w}$ the adjoint displacement field, $\bar{\mathbf{E}}$ the variational Green-Lagrange strain, and Γ_0^{ske} the boundary of the skeleton layer. For more details, the reader may refer to ref. [19].

People usually introduce a fixed directional spring at the end of the actuator to model the interaction, as in refs. [21,22]. When the actuator undergoes large deformation during grasping, the interaction forces need to keep in line with the normal of the contact surface. In this work, instead of relying on linear springs which generate reactive forces only along a fixed direction, we define uniform interaction pressure loads throughout the deformed configuration of the gripper as shown in Figure 3.

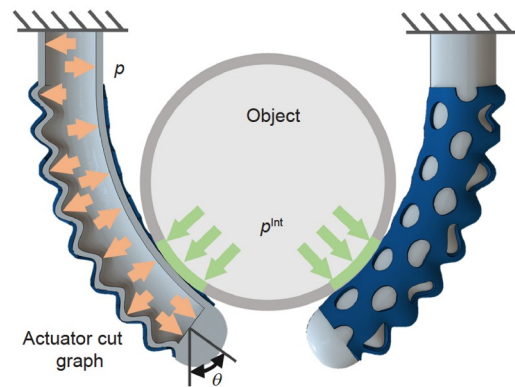


Figure 3 (Color online) The schematic diagram of a soft gripper: the gripper sustains actuation pressure in its interior air channel and distributed interaction loads from the grasped object.

3.2 Design and postprocessing-for-manufacture

During the design evolution, we choose the steepest descent algorithm to iteratively renew the skeleton profile until the optimization is converged. As shown in Figure 4, the iteration starts with two identical initial level-set profiles named as LS-I and LS-II which describe the top half and bottom half of the skeleton, respectively. The algorithm is gradient-based and thus the optimized result naturally depends on the initial guess. To prevent possible premature local optimal, different initial designs should be tried. As a common practice in level set based topology optimization, the initial profile is set to have dense holes, uniformly distributed inside the design domain and on the boundaries, to allow possible complex topological evolution. The nonlinear finite element analysis based on eq. (7) and sensitivity analysis eq. (11) are conducted to simultaneously update the two level set functions. The iteration process continues until the skeleton profile fulfills the predefined convergence criteria. The finite element model is generated for each optimization iteration according to the level-set profile which is realized by deactivating the corresponding elements belonging to the void phase.

In postprocessing for manufacture, we first triangulate the optimized design with the locally optimal delaunay-refinement method [23] to generate conformal and high-quality mesh in the planar domain. Thereafter, the coordinates of each vortex are mapped to the 3D space according to the relation in eqs. (4) and (5) while the connectivity remains the same. This post-processing generates the skeleton surface with complex topological characteristics, which reconstructs the optimization results while maintaining the boundaries' smoothness. We then refine and offset the meshes of skeleton

surface in the 3D space to generate the skeleton entity in the STL file format for additive manufacturing and compound molding.

4 Design for interaction

4.1 Numerical implementation

The material and geometry parameters used for skeleton design are summarized in Table 1. The material parameters are adopted from refs. [19,24]. The performance of actuator in fact depends on the material modulus and thickness ratio of skeleton to bladder. To address the compromise between deformation and stiffness requirement, we perform a dimensionless analysis and preliminary optimization trainings to refine the parameter range. We find that a skeleton layer too thin can provide limited stiffness to withstand interaction loads, while a skeleton layer too thick suppresses the deformation capability. A set of numerical studies show that the material modulus ratio falling between 30 and 40 and thickness ratio between 1/3 to 1/2 favor the optimal design.

We use Abaqus as a finite element solver. The other sections of the optimization are all carried out iteratively on our customized MATLAB program, including the evolution of the level set boundary, the sensitivity calculation, and the post-processing. To facilitate the general convergence of iterations and highlight the nonlinearity effect of the actuator deformation, we adopt a gradually loading scheme as reported in ref. [25]. The iteration starts with a relatively low pressure (say, 10 kPa) to induce a small strain which falls into the linear deformation region, and subsequently the linear optimization result is taken as an initial guess for the nonlinear optimization. While the pressure further increases

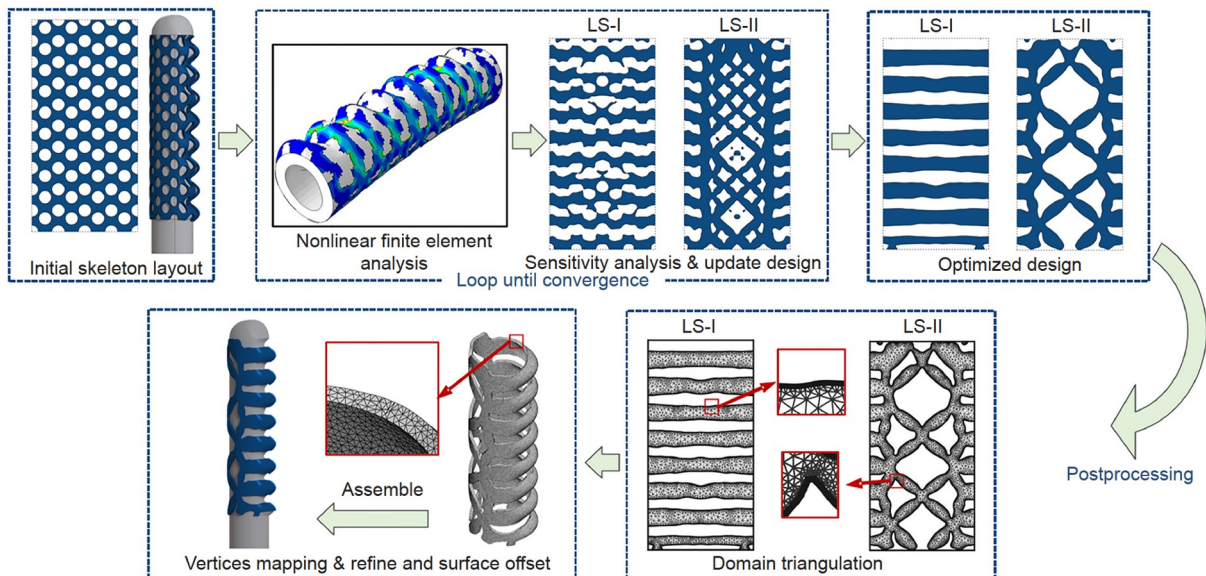


Figure 4 (Color online) Flowchart of designing and postprocessing the optimized skeleton layout.

Table 1 Material and geometry parameters for gripper design

Parameter	Value
Major axis a_1 (mm)	10
Major axis a_2 (mm)	10
Minor axis b (mm)	8
Actuator length l (mm)	80
Period of bellows T (mm)	20
Actuator thickness t^r (mm)	3
Amplitude controlling factor s	0.6
Skeleton thickness t^s (mm)	1.5
Shear modulus of rubber μ (kPa)	184
Bulk modulus of rubber κ (kPa)	18.35
Young's modulus of skeleton E (MPa)	19.46
Poisson's ratio of skeleton ν	0.495

up to 20 kPa, the structure undergoes large strain and the skeleton profile is further refined during this stage.

4.2 Effect of interactions on the optimal skeleton

To investigate how the interaction condition affects the skeleton layout, we set three typical load conditions as shown in Figure 5: (1) concentrated force at the fingertip; (2) uniform pressure along half length of the actuator; (3) uniform pressure along the actuator. For the sake of comparability, the resultant normal force is equal to 0.3 N for all load cases. The initial designs for all three cases are distributed with spaced dense circle holes. It is found that, for a given problem setting, as long as sufficient holes are uniformly placed in the initial design, the resulting skeleton layout exhibits the same topological pattern and very similar feature sizes, leading to almost the same performance.

The optimal skeleton patterns shown in Figure 5 (named as Skeleton A, Skeleton B, and Skeleton C from left to right) depend largely on the load condition and share common characteristics which are physically explainable as follows.

(1) The circumferential bars cover the peak area of the Pneu-Nets structure to constrain the radial inflation and enhance the axial extension of top half surface.

(2) The axial patterns in bottom half surface constrain the extension.

(3) The parallelogram mechanisms in bottom half surface expand along radial direction and shrink along axial direction upon pressurization.

(4) The major difference can be identified in the bottom half surface and it is strongly related to the predefined load condition. When the interaction loads move to the fingertip (from the condition in Figure 5(c) to that in Figure 5(a)), the optimized skeleton tends to reinforce the stiffness of the finger to bear larger bending moment, trading deformation for stiffness. For evenly distributed interaction loads, we can

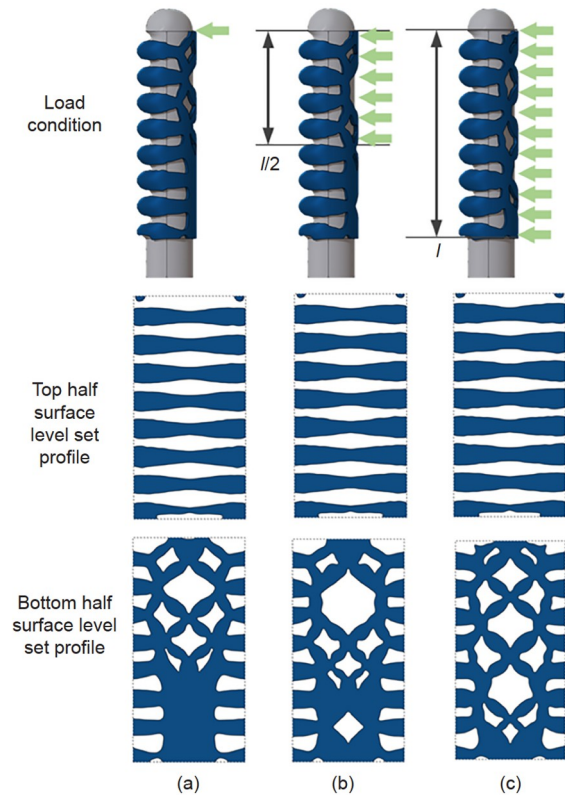


Figure 5 (Color online) Three different interaction loads and the corresponding optimal skeleton layouts. (a) Concentrated force at the fingertip; (b) uniform pressure along half-length; (c) full-length of the actuator.

observe a periodical arranged pattern in the bottom half surface.

5 Experiment: Fabrication, characterization, and demonstration

5.1 Fabrication

The soft bladder is cast with Dragon Skin 20 (Smooth-On, USA), a compound silicone mixed with Part A and B (1A:1B by volume or weight). After the silicone is sufficiently mixed, vacuum degassing is conducted to expel the air trapped in the liquid rubber. An injection pump is then employed to inject the uncured rubber into a mold, as shown in Figure 6(a), which is made by 3D printing to create the desired complex geometry. A release agent (Universal Mold Release, Smooth-On, USA) is applied on the surface prior to casting to prevent mold degradation. To avoid any air bubble left during the process, the injection spot is set on the base of the mold. The mold is then cured under 45°C for approximately 1 h, and the soft bladder is obtained after demolding. The skeleton layer is fabricated with HeiCast 8400 (Shore 90A, WeNext Technology, China) and the compound mold techniques are adopted.

We conduct experiments in order to evaluate the grasping

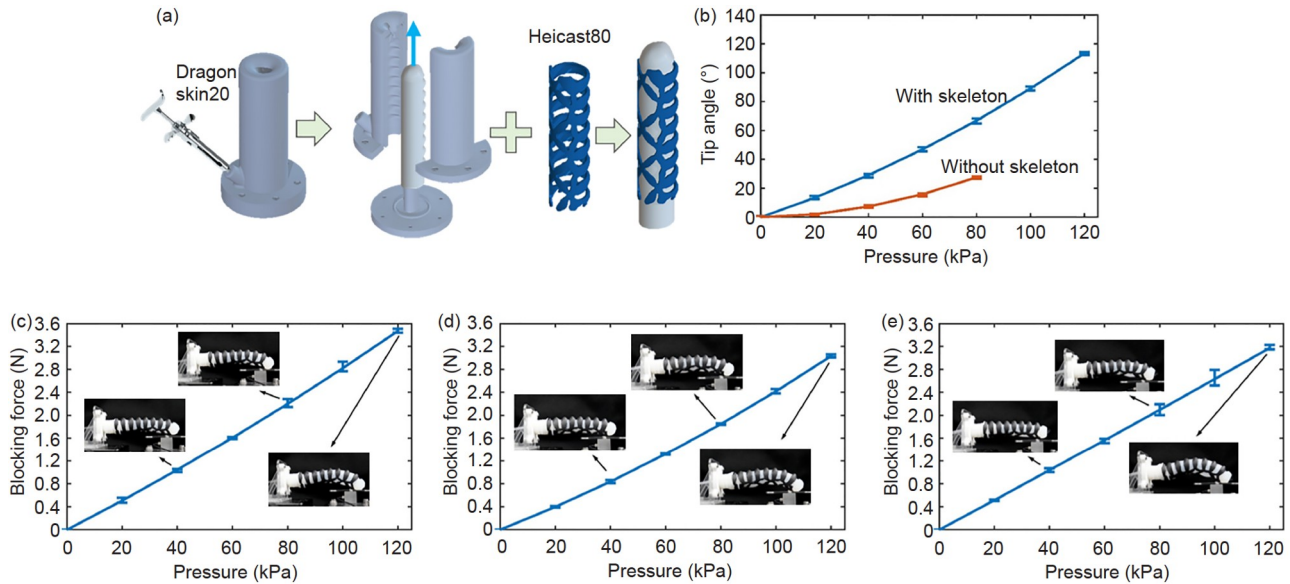


Figure 6 (Color online) (a) The actuator fabrication process: injection molding, rubber demolding and assembling; (b) the tip angle of the actuator with/without the skeleton; (c)–(e) blocking force tests with skeleton layout optimized under different conditions: (c) Skeleton A; (d) Skeleton B; (e) Skeleton C.

ability of GPN actuators and test their distinct capabilities, which are divided into two parts. First, the blocking force and free motion performance of each individual soft actuator are characterized. Second, with grippers consisting of PPNs covered with specially optimized skeleton, grasping and falling tests are conducted for comparison.

5.2 Actuators characterization

We first conduct free motion tests and it is found that, without interaction forces, skeletons A, B and C lead to very similar deformation characteristics. To demonstrate the enhancement of deformation abilities owing to the skeleton, the tip angle of the PPNs with Skeleton C is compared with its counterpart without skeleton design, as recorded and shown in Figure 6(b), and it can be concluded that the incorporation of skeleton design significantly improves the bending motion. It is noted that the actuator without skeleton is prone to rupture over 80 kPa.

We inflate three actuators with a quasi-static pressure, from 0 to 120 kPa, and measure their blocking forces by an S-beam load cell (ARIZON China). As shown in Figure 6(c)–(e), Skeleton A under this boundary condition performs the best, with the maximum force output of approximately 3.55 N.

5.3 Grasping and falling tests

As shown in Figure 7(a) and (b), the two grippers are assembled with PPN actuators consisting of Skeleton A and Skeleton C in Figure 5(a) and (c), respectively. To verify the effectiveness of the proposed method, a straightforward solution is to exchange the corresponding load condition of

design candidates shown in Figure 5. However, it is difficult to realize and measure the *in situ* load distribution in practice. Instead, we design a grasping and falling test to qualitatively evaluate the performance of PPN actuators. We first mimic the load condition of concentrated load by lifting objects in the vertical direction, denoted by Experiment A (Supporting Information). In this way, the weight of the object naturally concentrates at the tip of individual actuator. Second, to simulate the distributed loads along the actuator, the gripper holds the circular objects by the side, where the normal forces are more likely uniform to produce distributed friction forces to withstand the object weight, denoted by Experiment B (Supporting Information).

During these two experiments, the two grippers are inflated simultaneously by connecting to the same air inlet, and the initial pressure is set to be high enough to firmly hold the object of a certain weight. Then we decrease the actuation pressure quasi-statically till either of the two grippers cannot prevent the object any longer from falling, as shown in Figure 7(c) and (d). The reader may refer to the supplementary Video S1 for the complete releasing process. Here it is noted that, the actuator which accomplishes the same grasping task while requires a lower actuation pressure implies a better interaction performance.

In Experiments A and B, for both grippers, we record the critical minimal pressure required to hold two identical bottles that can change in weight by filling water, as shown in Figure 7(e) and (f). As expected, in Experiment A, the gripper equipped with Skeleton A outperforms that equipped with Skeleton C when the bottle weight is over 150 g. It is observed that Skeleton C outperforms A under 150 g, and we infer the discrepancy may originate from the deviation of

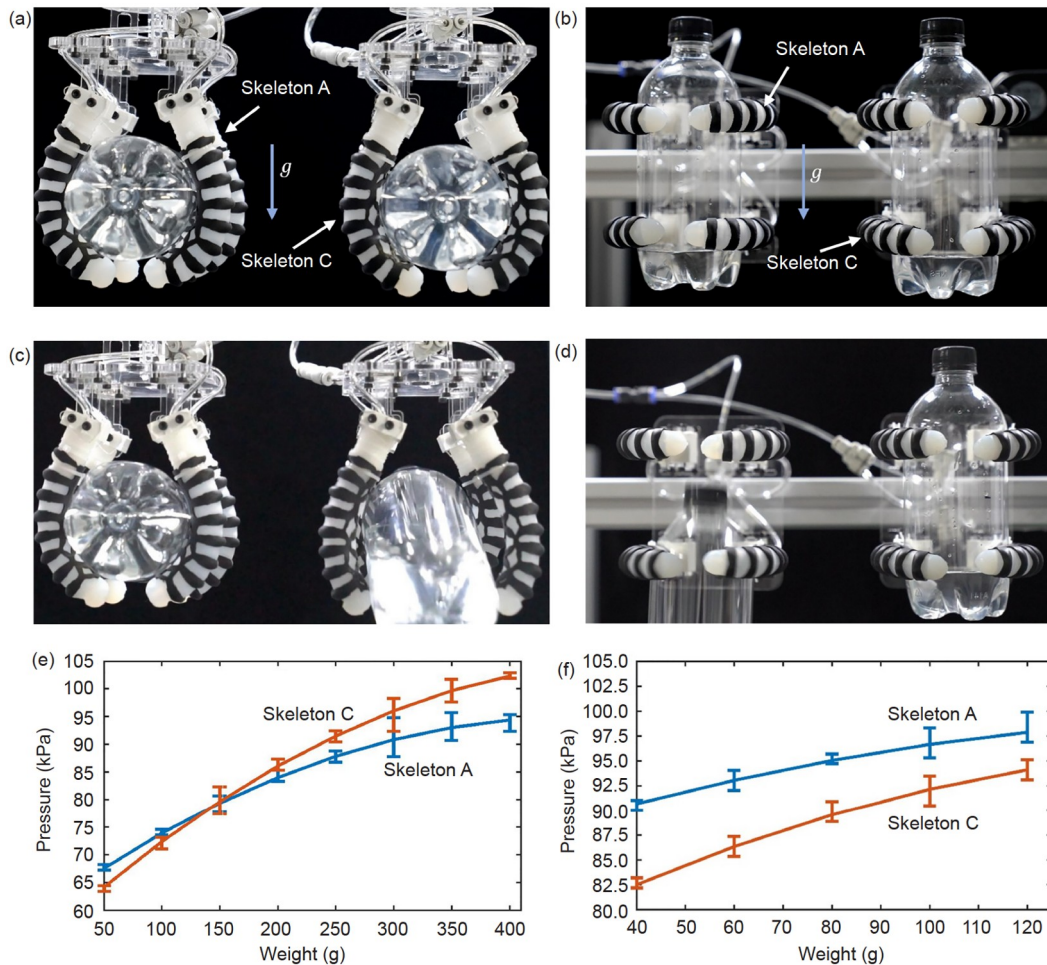


Figure 7 (Color online) The static states of grippers in (a) Experiment A and (b) Experiment B; (c), (d) objects' falling as the pressure decreases; the minimal pressure required to hold objects in (e) Experiment A and (f) Experiment B.

load condition from the optimization model. In Experiment B, the performance of gripper equipped with Skeleton C consistently exceeds that equipped with Skeleton A. This experiment qualitatively verifies the superiority of the skeleton and validates the effectiveness of the proposed design framework.

6 Conclusion and future work

In this work, we propose the automatic design and manufacture of soft pneumatic grippers, with a focus on the incorporation of interaction scenarios into the design model and the aim to reach the optimal compromise between conformability and stiffness. The design paradigm rests on optimizing the skeleton layer that is attached to a parametrized Pneu-Nets actuator to program the resultant motion behavior and interaction performance. The skeleton layout is subject to topology optimization under different load conditions to simulate power grip or pinch grip, and it is found that the patterns of the skeleton are physically explainable and highly

relevant to the interaction conditions. We demonstrate a grasping and falling test for the two optimized fingers under the specified grasping postures to mimic the predefined loads. The critical minimal actuation pressure of object's falling proves the superiority of each design under its predefined interaction scenario, which reveals the importance of considering interaction performance when designing soft robots.

Although we mainly focus on the skeleton design to imbue the soft rubber with better interaction performance, it is acknowledged that channel geometry plays an equally important role. The design method can be further extended to concurrent geometry, actuation, and material optimization to explore the design space. Besides, we hope to capture the interaction condition more accurately in practical grasping scenarios, e.g., proprioception by distributed sensing and data-driven methods, and incorporate it into the optimization model to enable novel soft grippers with optimal interaction performance in a general sense. In addition to the design aspect, for a given task with known target objects, a conformal contact also requires well-organized grasp planning

and control algorithms working in concert with the gripper's physical characteristics, especially for multi-fingered grippers.

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Supporting Information

The supporting information is available online at tech.scichina.com and link.springer.com. The supporting materials are published as submitted, without typesetting or editing. The responsibility for scientific accuracy and content remains entirely with the authors.

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